

GROUP 1

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Table of Content

1. Introduction 2

2. Accommodation Market Growth and Challenges 2

2.1. Project Objective 4

3. Methodology 4

3.1. Dataset Overview 5

3.2. Exploratory Data Analysis and Preparation 5

4. Machine Learning Approach 9

4.1. Listing-Profiles 10

4.2. Pricing Analysis 12

5. Limitations, Results and Business Insights 13

6. Conclusions and Recommendations For the Future 15

References 18

Bibliography 22

1. Introduction

Airbnb connects 5.5 million hosts with over 2.5 billion guests across 9 million listings worldwide, creating authentic local experiences globally (Airbnb, 2026).

This project analyses Airbnb New York City (NYC) listing data (Dgomonov, 2019) for listing segmentation and matching price predictions, following a classical machine learning (ML) workflow (Géron, 2025). The analysis covers data understanding, preparation and feature engineering, modelling and evaluation. The resulting insights aim to inform pricing strategies, targeted host recommendations and marketplace positioning.

2. Accommodation Market Growth and Challenges

The online accommodation market is growing steadily, with short-term rentals and homestays expanding at more than twice the overall market rate, and Airbnb holding its largest share (Figure 1).

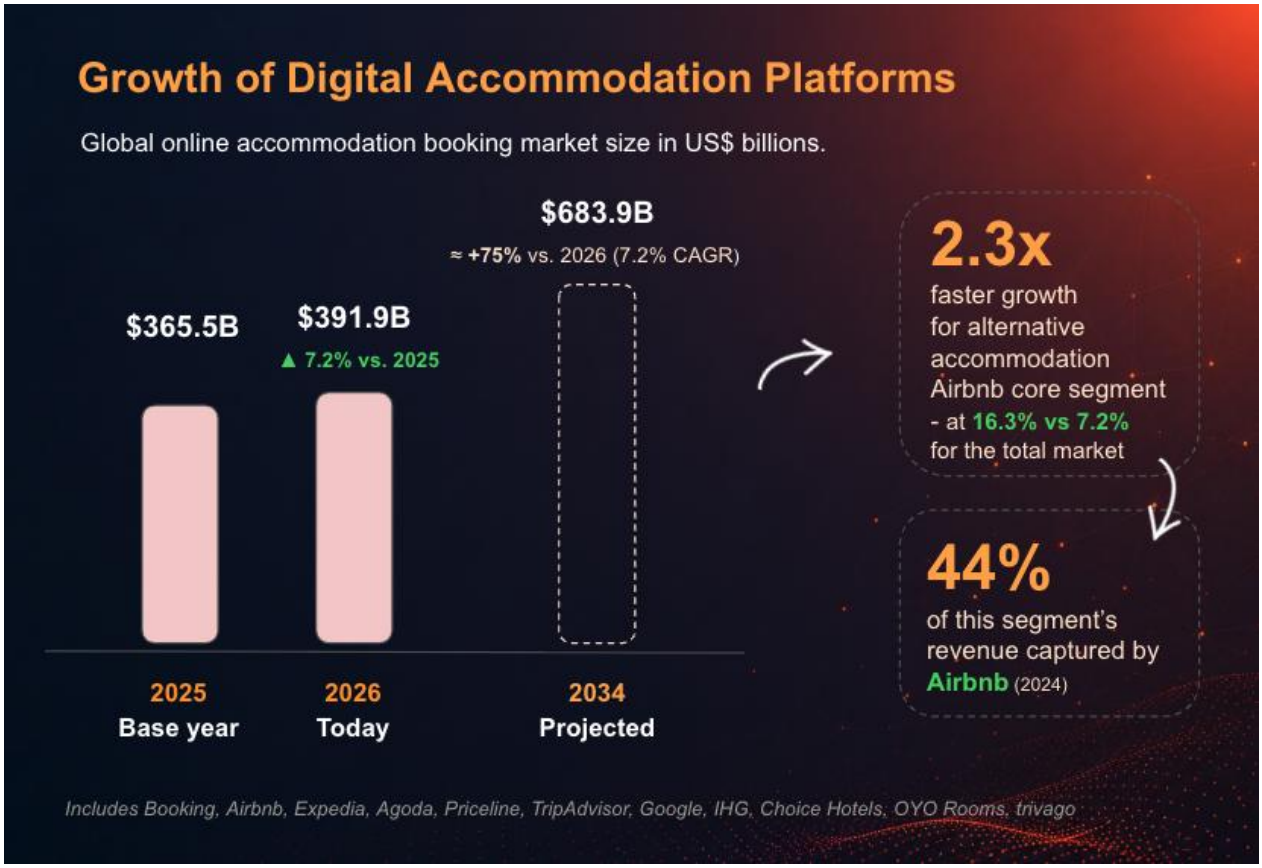


Figure 1. Growth of Digital Accommodation Platforms. Source: Developed by the authors, based on the insights from Schaal (2025) and Fortune Business Insights (2026a, 2026b).

The increasing adoption of ML enables more data-driven strategies, intensifying competition. As supply surges, the challenge shifts from scarcity to effective discovery, aiming to reduce information and choice overload (Athi Karthick, Majumdar and Bose, 2025).

Pricing remains the primary driver of booking decisions (Expedia Group, 2025), bringing data-driven pricing tools and personalised recommendations to the forefront (Statista and Booking, 2025). Airbnb already applies ML–including clustering and regression–to

address multidimensional pricing challenges and host profiling (Wang and Nicolau, 2017; Rezazadeh Kalehbasti, Nikolenko and Rezaei, 2021; Li, Liao and Gao, 2022; Zhang, 2025).

2.1. Project Objective

This analysis aims to identify distinct Airbnb listing profiles and evaluate their contribution to price prediction, driven by the following business analytical question:

What distinct listing profiles can be identified, and can these profiles improve price prediction?

3. Methodology

The project follows a classical ML workflow (Géron, 2025), as summarised in Figure 2. Findings are translated into business insights and actionable recommendations.

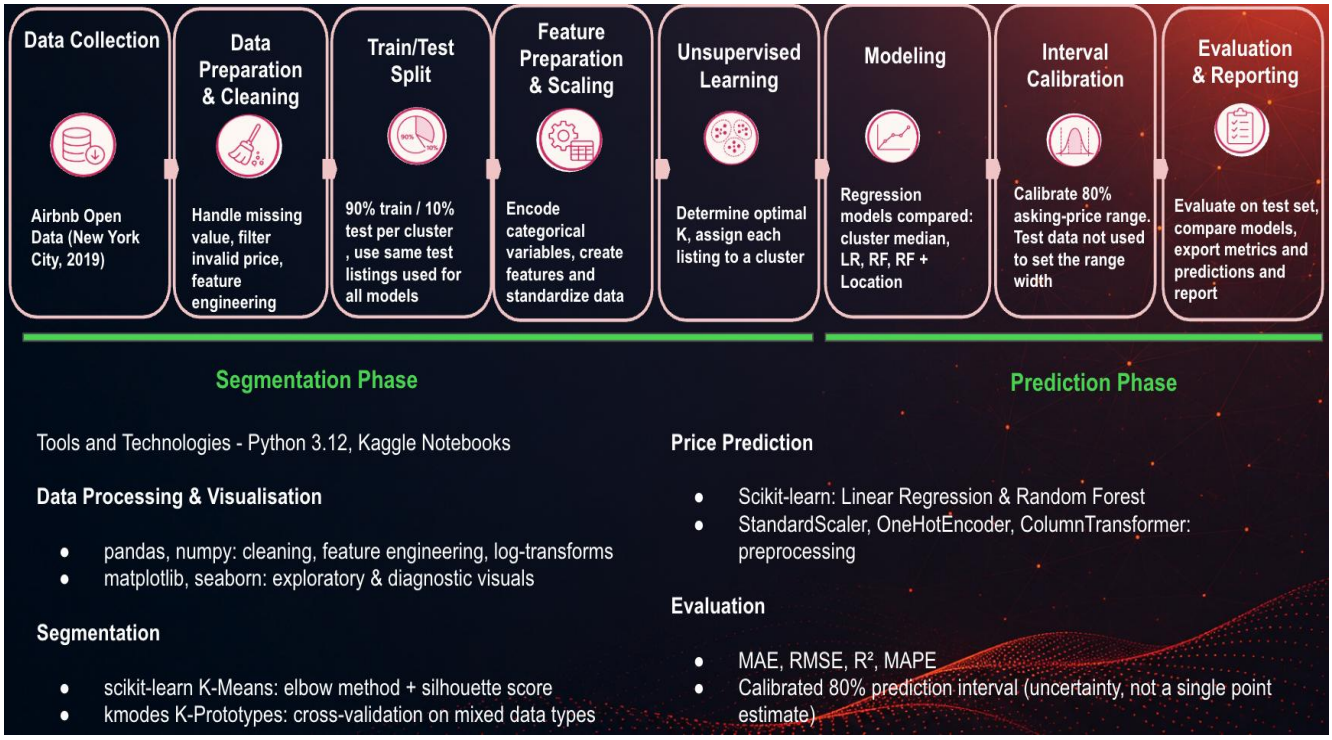


Figure 2. Analytical Pipeline and tooling used for Airbnb Segmentation and Price Prediction. Source: Authors' own work.

3.1. Dataset Overview

The NYC Airbnb dataset (Dgomonov, 2019) used in this analysis contains 16 listing and host attributes across 48,895 observations (Figure 3).

Dataset content: information on Airbnb listings in New York City during 2019, including geographical characteristics, room types, pricing, availability, review activity, and host-related attributes.

48895 records, **16** attributes

Variable	Description	Business Relevance
id	Unique identifier for each listing	Identifies individual properties
name	Listing name	Descriptive information about the property
host_id	Unique identifier for each host	Enables host-level analysis
host_name	Host name	Host information
neighbourhood_group	NYC borough	Geographic market segmentation
neighbourhood	Specific neighbourhood	Local market differences
latitude	Listing latitude	Location-based analysis
longitude	Listing longitude	Location-based analysis
room_type	Type of accommodation	Defines customer offering categories
price	Nightly price	Represents market positioning
minimum_nights	Minimum stay requirement	Indicates booking strategy
number_of_reviews	Total reviews received	Proxy for customer engagement
last_review	Date of latest review	Indicates recent activity
reviews_per_month	Review frequency	Proxy for listing activity
calculated_host_listings_count	Number of listings managed by host	Indicates host profile
availability_365	Available days per year	Represents supply availability

Figure 3. Dataset overview. Source: Authors' own work.

3.2. Exploratory Data Analysis and Preparation

Exploratory data analysis (EDA) focused on data quality, missingness, distributions, and potential outliers to guide preprocessing decisions (Géron, 2025). Missing values were concentrated in `last_review` and `reviews_per_month`, affecting approximately 20% of listings (Figure 4).

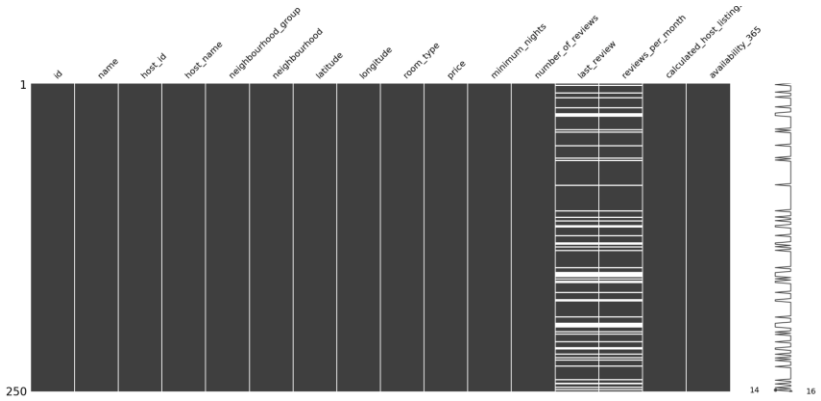


Figure 4. Airbnb missing data visualisation. Source: de Jonge-Gryciuk et al. (2026)

Numerical variables had weak linear relationships with price (Figure 5), with slightly stronger relationships evident for `room_type` and location features. Manhattan showed the highest median price ($150), while entire homes/apartments had a higher median ($160) than private ($70) and shared rooms ($45) (Figure 6), and overall median price at $106. Price and `minimum_nights` were strongly right-skewed, with only 11 $0 price listings.

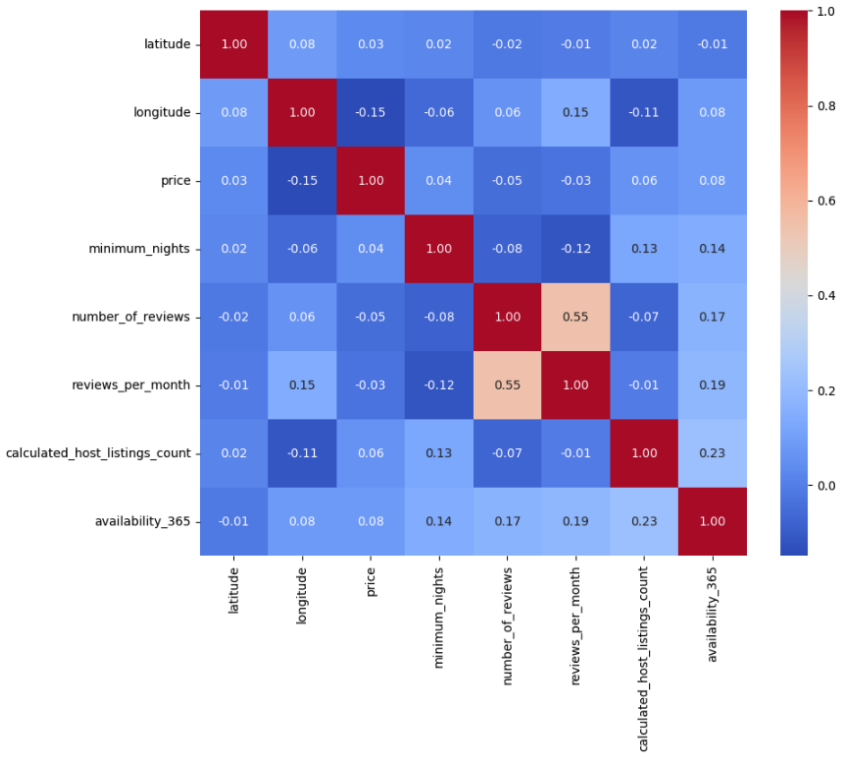


Figure 5. Price correlation matrix. Source: de Jonge-Gryciuk et al. (2026)

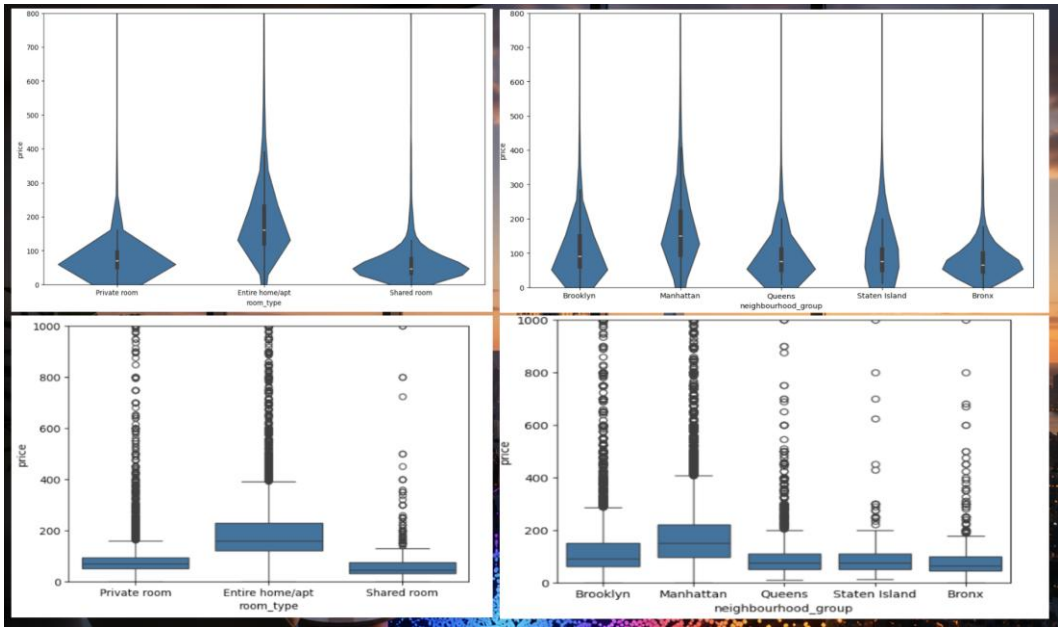


Figure 6. Violin and boxplots for various Airbnb features. Source: de Jonge-Gryciuk et al. (2026)

Preprocessing followed these findings and Géron's (2025) suggested steps. Missing `reviews_per_month` values, 100% overlapping with zero reviews, were replaced with 0; a binary `has_reviews` variable was created; zero-price listings were removed, while high-price observations were retained; highly skewed variables were transformed using `log1p` to reduce the influence of extreme values in distance-based clustering (Figure 7). Identifiers, personal text, `last_review`, and high-cardinality `neighbourhood` were excluded, while `latitude` and `longitude` represent location.

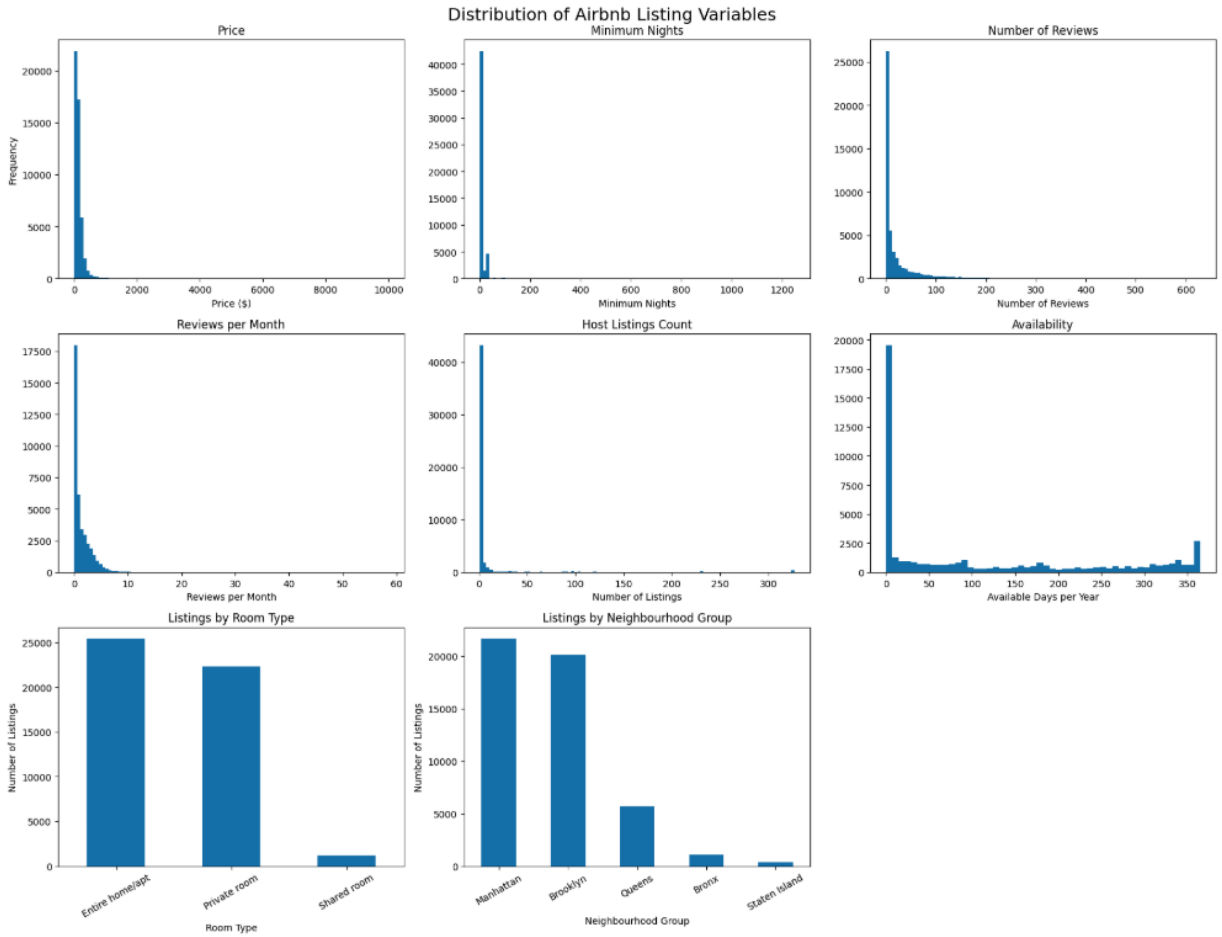


Figure 7. Airbnb various feature distributions. Source: de Jonge-Gryciuk et al. (2026)

Numerical features were standardised to prevent scale differences from dominating distance calculations. `room_type` was one-hot encoded for K-Means and retained categorically for K-Prototypes.

4. Machine Learning Approach

This analysis used clustering and regression, with model selection guided by the business objective, data characteristics, and implementation trade-offs.

Unsupervised clustering was used to identify listing profiles. K-Means and K-Prototypes were evaluated independently to determine the optimal number of clusters and compared for composition and business meaningfulness (Figure 8).

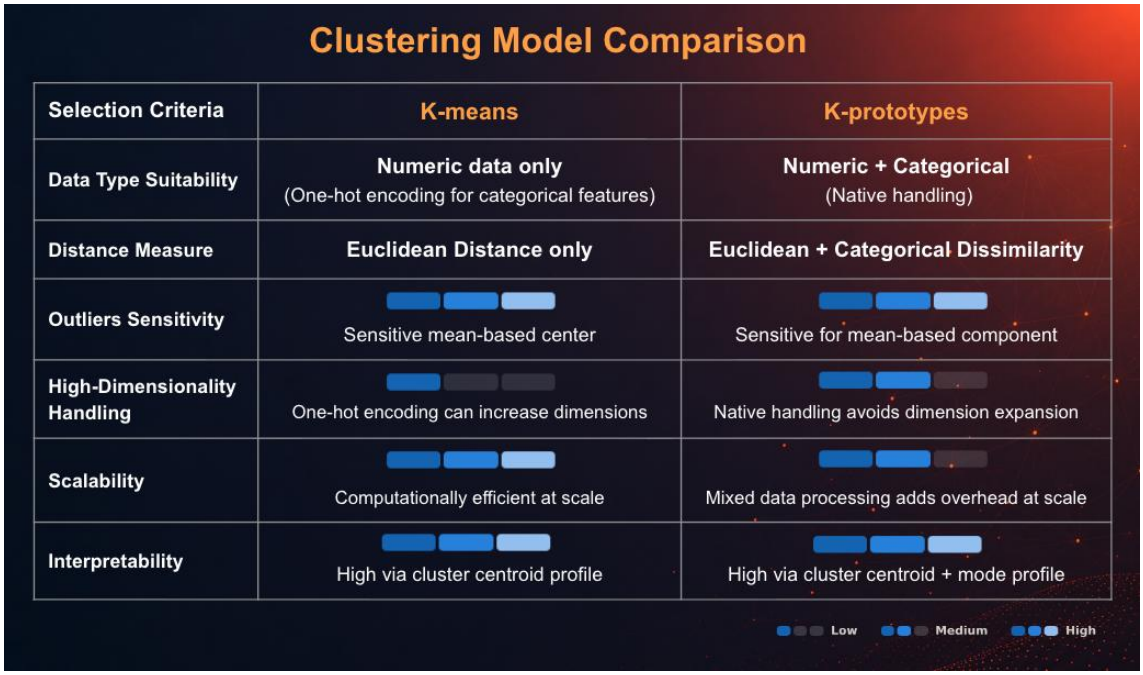


Figure 8. Comparison of Clustering Models. Source: Developed by the authors, based on Huang (1980); Aggarwal and Reddy (2014); Costa, Papatsouma and Markos (2023); Jain et al. (2026).

Regression modelling compared Linear Regression (LR) and Random Forest (RF) models across cluster-specific and global approaches (Figures 9, 11), assessing whether segmentation enhances price prediction.

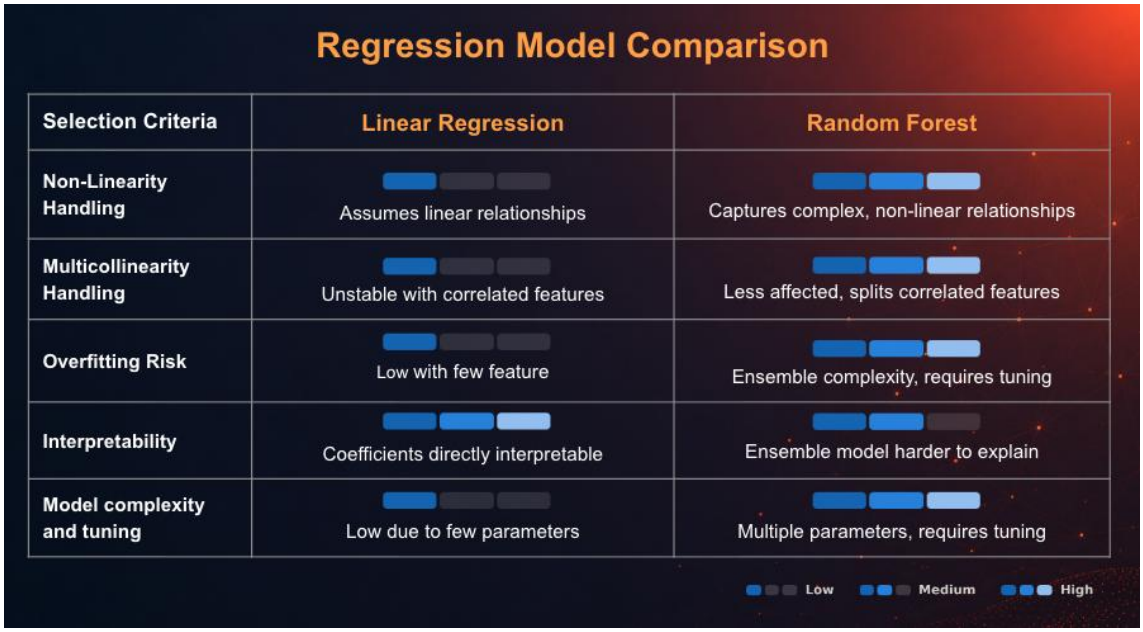


Figure 9. Comparison of Regression Models. Source: Developed by the authors, based on Hastie, Tibshirani and Friedman (2009)

4.1. Listing-Profiles

K-Means can identify meaningful groups based on shared characteristics (Pant *et al.,* 2025), which may reveal distinct Airbnb listing profiles and inform pricing strategies.

The Elbow Method and Silhouette Score supported the K-selection (Li, Liao and Gao, 2022). At K = 3, K-Means and K-Prototypes produced almost identical cluster profiles despite handling categorical features differently, strengthening confidence in the robustness of identified patterns.

Cluster profiling using medians for numerical features and the leading category for categorical features revealed three distinct listing profiles (Figure 10). These were

subsequently evaluated through regression modelling to determine whether segmentation improves price prediction.

Clusters Identified via K-Means and K-Prototype:

Cluster 0: 26,507 listings · median price $110 · 3 minimum nights · 1 host listing · 0 days availability · most commonly entire home/apt · Manhattan

Cluster 1: 4,482 listings · median price $149 · 30 minimum nights · 18 host listings · 319 days availability · most commonly entire home/apt · Manhattan

Cluster 2: 17,895 listings · median price $100 · 2 minimum nights · 1 host listing · 160 days availability · most commonly private room · Brooklyn

Cluster	Listings	Median Price	Min. Nights	Reviews	Host Listings	Availability	Leading Room Type	Leading Borough
0	26,507	$110	3	2	1	0	Entire home (53.0%)	Manhattan (45.6%)
1	4,482	$149	30	1	18	319	Entire home (67.8%)	Manhattan (72.49%)
2	17,895	$100	2	36	1	160	Private room (50.7%)	Brooklyn (42.16)

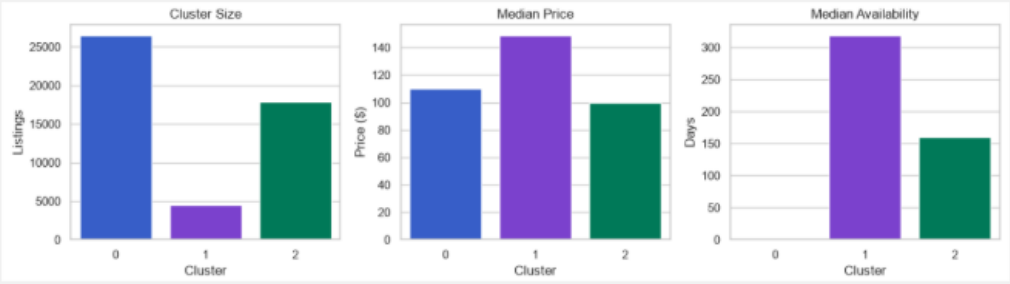


Figure 10. Clusters, with their size, median statistics and rates. Source: Authors' own work.

4.2. Pricing Analysis

The selected experimentation approach (Figure 11) examined whether the identified clusters contribute to price prediction. Two regression models were compared: LR and RF, while the contribution of additional location features was

evaluated. Model performance was evaluated using Mean Absolute Error (MEA), Root Mean Squared Error (RMSE), and R-Squared (R^2). Price was excluded from clustering to ensure segments were identified independently of the subsequent prediction target.

Results show that RF performs better than LR. Adding location features further improves prediction. However, modelling all listings together, without cluster separation, produces the best price-prediction results.

Experiment	Model	Cluster used for Modelling?	Location used?	MAE ($) ↓	RMSE ($) ↓	R^2 (↑)
1 Within clusters (baseline)	Cluster Median (baseline)	Yes (median baseline per cluster)	No	85.06	109.63	-0.025
2 Within clusters	Linear Regression	Yes (separate model per cluster)	No	64.23	86.21	0.076
3 Within clusters	Random Forest	Yes (separate model per cluster)	No	58.32	79.96	0.132
4 Within clusters	Random Forest	Yes (separate model per cluster)	Yes (Borough + Neighbourhood)	57.88	79.41	0.136
5 Without separate cluster models (global model)	Random Forest	No (one model for all listings)	Yes (Borough + Neighbourhood)	**57.72**	**79.13**	**0.134**

What was tested?

1 Baseline established with the median price of each K-Means cluster compared with the actual price.

1 → 4 Models trained separately within each of the 3 K-Means clusters.

5 One global model trained on all data, ignoring clusters.

Figure 11. Price prediction model experimentation and results. Source: Authors' own work.

Since the best-performing model explains only 13.4% of price variation (Figure 11), we calibrated an 80% typical asking-price range around the model's prediction rather than relying on an exact predicted price. On the held-out test set, 79.4% of actual prices fell within these ranges, with an average range width of approximately $131. These ranges reflect observed listing prices, not commercial effectiveness.

Full analysis is available on Kaggle (de Jonge-Gryciuk *et al.*, 2026).

5. Limitations, Results and Business Insights

The project aimed to identify host segments and evaluate their contribution to pricing strategies. This was only partly achieved due to limitations of the NYC dataset (Dgomonov, 2019). Weak price relationships (Figure 5) and the best model explaining only 13.4% of price variation (Figure 11) suggest important data gaps, particularly guest, property, demand and commercial-performance features absent from the predominantly supply-side dataset (Rezazadeh Kalehbasti, Nikolenko and Rezaei, 2021; Cheng *et al.*, 2026; Culotta *et al.*, 2026). Nevertheless, the resulting personas (Figure 12) remain valuable for targeted host strategies. Substantial price variation within segments (Figure 13) further supports listing-level asking-price ranges rather than segment-based pricing. Generalisability is also limited by the NYC-only scope, while reviews may suffer from selection bias (Han and Anderson, 2026).



Figure 12. Data-driven host personas derived from the clustering analysis. Source: Developed by the author, drawing on the persona development approach discussed by Prochner and Godin (2025).

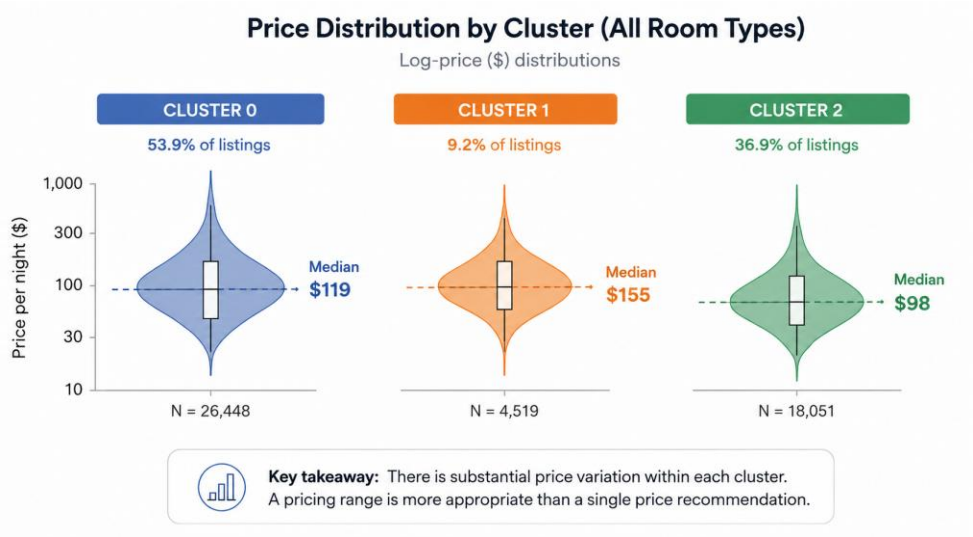


Figure 13. Price variation per cluster based on clustering analysis. Authors' own work.

6. Conclusions and Recommendations For the Future

Future research should enrich the dataset with property and contextual features, including size, rooms, amenities, accessibility and neighbourhood characteristics (Culotta *et al.*, 2026). Unstructured data could further capture missing attributes; review sentiment may provide richer insights into guest experience (Rezazadeh Kalehbasti, Nikolenko and Rezaei, 2021), while listing descriptions and visuals could reveal perceived quality, space or decor (Cheng *et al.,* 2026). Integrating these features with demand data could strengthen pricing models and enable genuine guest-level segmentation (Wu and Schmierer, 2026). The identified host personas could inform tailored engagement, benchmarking and reactivation strategies (Michelli, 2020; Kumar, 2025; Prochner and Godin, 2025). Finally, replicating the analysis across multiple markets is necessary to assess whether these findings generalise beyond NYC (Cai, Namkoong and Yadlowsky, 2026).

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